Sven Amberg · Student and Tech and Space Enthusiast · Trondheim, Norway · **sven.amberg@outlook.com** · +47 478 011 04

Sven | Amberg

- **Status:** Logistics, M.Tech. Electronic Systemdesign, Electronics Development
- **Fields:** Project Management, PCB Design, System Engineering, Machine Learning

Summary

Motivated student specializing in Electronic System Design and Space Systems with experience in PCB design, systems engineering, and electronic design using GNU Radio, Altium, and KiCad. Proven ability to lead projects from concept to product, including satellite design and contributing to the Solar World Challenge. Skilled in SysML, IoT, and Python, with strong team collaboration experience. A creative problem-solver with 15+ years of saxophone playing, and a passion for volleyball, musicals, video games, books, and Sci-Fi.

- **Tech:** Python, GNU Radio, Altium, KiCad, Git, Tensorflow, C++, MatLab, Verilog, HDL
- **Loves:** Global Game Jam, Sci-Fi and Fantasy series, Playing the Saxophone, Musicals, Working on DIY Projects

Education

2020-2026 - Norwegian University of Science and Technology

Integrated Master Electronic Systemdesign and Innovation

- Spesialising in Space Systems
- Classes covered project development, IoT design, signal processing, GNSS, electronics and system design. Also developed Ultrasound sensor system for the Blind

2023 - 2024 - Technical University of Berlin

Year Abroad

- Mastered twelve months of study and trans-cultural experience in Berlin, Germany
- Finished machine learning, speech technology, space engineering, satellite operations and development. Also helped bring an Exoskeleton (RISE EXO) to life for the Cybathlon Challenge 24

Experience

2025 / 09 - 2026 / 01 - ESA ESOC

OPS-SAT Space Lab Intern

- Developing precise orbit determination algorithm using GNSS data
- Supporting experimenters, setting up, developing and carry out experiments on satellites

Supporting satellite design and development through CDR and IRRSettung up testbenches for OPS-SAT satellites

2024 / 08 - NTNU

Student Assistant

- Organise and plan Activities for International Master Students
- Planning, budgeting and contact relations

2023 - Summer - Space Norway

Summer Engineering Intern

- Research and simulate the possibility of using in-house Satellites as Ship SOS Relays
- Using GNU Radio, Python and ML to simulate ship SOS and the satellites movement and equipment

2021 June - 2021 December - NTNU

Student Assistand

- Creatid new learning framework and learning guide for ESP32 and small electronics projects
- Worked as an assistant and helped students develop their projects

Extracurricular

2020 - 2023 - ORBIT NTNU
CFO and Electronics Engineer

- Led the key accounts team and handle all the finances with a budget over 1 mill NOK under the launch of SelfieSat. Designed PCB for BioSat and worked on electronics design, soldering and cabling.
- Led the electronics design for the Sub-Orbital team 2020. Built the Payload for Propulse NTNU. It won 1.Place.

2024 - 2025 - Nordlys NTNU
Logistics

- Planning the Competition logistical means, contacting sponsors and shipping the Car
- Contributing to the healthy and successfull participation at the World Solar Challenge 25 in Australia

2024 - Present - ARK - Akademisk Radio Klubb
Amateure Radio Member

- Small projects, radio contests, building groundstations and morse code recogniser software.
- Working on radios, antennas and helping around with DIY projects

2024 - 2025 - Student Parliament NTNU
Representative

- Student representative in NTNU Student Parliament
- Representing over 40 000 students and helping decide NTNUs future

2020 - 2024 - Omegarevyen
Revy Kose Boss and member

- Koseboss one year. Helping with the Board, arranging social activities and leading a team
- Helping the revy as a normal member with everything from PR, social and backstage

2012 - Present - Many different orchestras and bands
Saxophone Player

- Performed at various events and concerts in Trondheim, other cities and in Germany.
- Contributed to arranging and organizing rehearsals and public performances. Started a student orchestra in Berlin

Publications

2024 - International Journal for Students as Partners
Design Thinking and Shared Ignorance as a Framework for Student-Faculty Partnerships

- Co-authored case study of the creation of teaching activities for wicked problems
- Worked together with students and professors, and studied how chared ignorance helped the cooperation between faculty and students.

Courses

2025 - Nord University
Space Hackathon Tekna

- Case hackathon in space debris
- WOrked in an international team creating a solution for protecting against and removing space debris

2024 - University of Brest
BIP Sustainable development, cooperation and design thinking

- Completed blended intensive course
- Techniques to solve real-world problems, design thinking and cooperation.

2024 - University of Split and Cadiz
BIP Computational Mathematics

- In-depth understanding of Computational Mathematics
- How to use Machine Learning in solving PDEs and FEM.

2024 - Krakow University
Enhance Hackathon about Smart Cities

- Case hackathon in development of smart cities
- WOrked in an international team creating a sustainable smart city solution for the future

2021 - NTNU
LAOS

- Course about teaching and understanding students and the classroom
- Needed to be able to work for the University. Learning how to teach, connect and understand students and give constructive and good feedback